Assignment 4

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Please note that

- **Released date: 30th June, Fri.**
- **Due date: 7th July, Fri., by 11:59pm.**
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like "120010XXX ASS1.pdf" (Your student ID + ASS No.). You may either typeset you answers directly using computers, or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size 10MB.)

Question 1 . Determine whether the following sets are bases for $\mathbb{R}^3$.

(a)
$$\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\end{bmatrix}, \begin{bmatrix}1\\1\\1\end{bmatrix}$$

(b)
$$\begin{bmatrix}1\\0\\1\end{bmatrix}, \begin{bmatrix}0\\0\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\end{bmatrix}$$

(c)
$$\begin{bmatrix}1\\0\\-2\end{bmatrix}, \begin{bmatrix}3\\2\\-4\end{bmatrix}, \begin{bmatrix}-3\\-5\\1\end{bmatrix}$$

Solution

(a) It is obvious that these 3 vectors are linearly independent and can span $\mathbb{R}^3$, so this set of vectors is a basis for $\mathbb{R}^3$.

(b) Because of the existence of 0 vector, these 3 vectors are not linearly independent, so this set of vectors is not a basis for $\mathbb{R}^3$.

(c) We can perform Gaussian Elimination on matrix get:

$$\begin{bmatrix}1&3&-3\\0&2&-5\\-2&-4&1\end{bmatrix} \to \begin{bmatrix}1&3&-3\\0&2&-5\\0&0&0\end{bmatrix}$$

It only has 2 pivot positions, so it doesn't form a basis for $\mathbb{R}^3$.

Question 2. Find a maximal linearly independent subset of the following set:

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \\ -3 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ -6 \\ 1 \\ -3 \end{bmatrix}, \begin{bmatrix} -4 \\ 3 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -5 \\ 9 \\ -4 \\ -7 \end{bmatrix} \right\}.$$

Solution

Assume $a_1 \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \end{bmatrix} + a_2 \begin{bmatrix} -2 \\ 3 \\ -3 \\ 4 \end{bmatrix} + a_3 \begin{bmatrix} 3 \\ -6 \\ 1 \\ -3 \end{bmatrix} + a_4 \begin{bmatrix} -4 \\ 3 \\ 1 \\ 2 \end{bmatrix} + a_5 \begin{bmatrix} -5 \\ 9 \\ -4 \\ -7 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$, then the augmented matrix is

$$\left[\begin{array}{rrrrr|c} 1 & -2 & 3 & -4 & -5 & 0 \\ 0 & 3 & -6 & 3 & 9 & 0 \\ -2 & -3 & 1 & 1 & -4 & 0 \\ 1 & 4 & -3 & 2 & -7 & 0 \end{array}\right] \rightarrow \left[\begin{array}{rrrrr|c} 1 & 0 & 0 & -2 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{array}\right].$$

Here we can see there are only 4 pivots in column 1, 2, 3 and 5, the column 4 is a free variable. So the maximal linearly independent subset for the set S is:

$$S_i = \left\{ \begin{bmatrix} 1 \\ 0 \\ -2 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 3 \\ -3 \\ 4 \end{bmatrix}, \begin{bmatrix} 3 \\ -6 \\ 1 \\ -3 \end{bmatrix}, \begin{bmatrix} -5 \\ 9 \\ -4 \\ -7 \end{bmatrix} \right\}.$$

Question 3. Given

$$\mathbf{x}_1 = \begin{bmatrix} -1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{x}_2 = \begin{bmatrix} 3 \\ 4 \\ 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 2 \\ 6 \\ 6 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} -9 \\ -2 \\ 5 \end{bmatrix}.$$

(a) Is $\mathbf{x} \in \text{Span}(\mathbf{x}_1, \mathbf{x}_2)$? why?

(b) Is $\mathbf{y} \in \text{Span}(\mathbf{x}_1, \mathbf{x}_2)$? why?

Consider whether the linear system $a_1\mathbf{x}_1 + a_2\mathbf{x}_2 = \mathbf{x}$ and $b_1\mathbf{x}_1 + b_2\mathbf{x}_2 = \mathbf{y}$ have solution. If not, then the vectors are linearly independent. Otherwise, the vectors are linearly dependent.

(a) Assume $a_1\mathbf{x}_1 + a_2\mathbf{x}_2 = \mathbf{x}$, write the linear equation into matrix format:

$$\left[\begin{array}{cc|c} -1 & 3 & 2 \\ 2 & 4 & 6 \\ 3 & 2 & 6 \end{array}\right] \to \left[\begin{array}{cc|c} -1 & 3 & 2 \\ 0 & 10 & 10 \\ 0 & 11 & 12 \end{array}\right] \to \left[\begin{array}{cc|c} -1 & 3 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{array}\right].$$

The linear system has no solution, thus $\mathbf{x} \notin \text{Span}(\mathbf{x}_1, \mathbf{x}_2)$;

(b) Assume $b_1\mathbf{x}_1 + b_2\mathbf{x}_2 = \mathbf{y}$, then

$$\left[\begin{array}{cc|c} -1 & 3 & -9 \\ 2 & 4 & -2 \\ 3 & 2 & 5 \end{array}\right] \to \left[\begin{array}{cc|c} -1 & 3 & -9 \\ 0 & 10 & -20 \\ 0 & 11 & -22 \end{array}\right] \to \left[\begin{array}{cc|c} -1 & 3 & -9 \\ 0 & 1 & -2 \\ 0 & 0 & 0 \end{array}\right].$$

The linear system has a unique solution: $b_1 = 3$, $b_2 = -2$, thus $\mathbf{y} \in \text{Span}(\mathbf{x}_1, \mathbf{x}_2)$.

Question 4. Determine if the set $S = \{(x, y, z)^T | xy = z^2\}$ is a subspace of $\mathbb{R}^3$.

Solution

Let $\mathbf{u}_1 = (x_1, y_1, z_1)^T$ and $\mathbf{u}_2 = (x_2, y_2, z_2)^T$, $\mathbf{u}_1, \mathbf{u}_2 \in S$.

$$\mathbf{u}_1 + \mathbf{u}_2 = (x_1 + x_2, y_1 + y_2, z_1 + z_2)^T$$
$$(x_1 + x_2)(y_1 + y_2) = x_1y_1 + x_2y_1 + x_1y_2 + x_2y_2$$
$$(z_1 + z_2)^2 = z_1^2 + 2z_1z_2 + z_2^2 = x_1y_1 + 2z_1z_2 + x_2y_2$$
$$(x_1 + x_2)(y_1 + y_2) \neq (z_1 + z_2)^2$$

Thus, S is not a subspace of $\mathbb{R}^3$.

Using examples for proof is also acceptable.

Question 5. Find the basis for

$$V = \text{Span}\left(\begin{bmatrix}1\\0\\-3\\3\\2\end{bmatrix}, \begin{bmatrix}3\\1\\0\\4\\0\end{bmatrix}, \begin{bmatrix}1\\1\\6\\-2\\-4\end{bmatrix}, \begin{bmatrix}3\\0\\-1\\1\\2\end{bmatrix}\right).$$

Solution

Using elementary row operations, rewrite $\boldsymbol{A}$ in row-echelon form as follows:

$$\boldsymbol{B} = \begin{bmatrix}1 & 0 & -2 & 0\\0 & 1 & 1 & 0\\0 & 0 & 0 & 1\\0 & 0 & 0 & 0\\0 & 0 & 0 & 0\end{bmatrix}$$

We can conclude that the basis for $\boldsymbol{A}$ is

$$w_1 = \begin{bmatrix}1\\0\\-3\\3\\2\end{bmatrix}, \quad w_2 = \begin{bmatrix}3\\1\\0\\4\\0\end{bmatrix}, \quad w_3 = \begin{bmatrix}3\\0\\-1\\1\\2\end{bmatrix}.$$

Question 6. Let

$$V = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} \in \mathbb{R}^4 \middle| x_1 + x_2 + x_3 + x_4 = 0 \right\}.$$

By the definition of basis, show that

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} \right\}$$

is a basis for V. Also find out the dimension of V.

Solution

Let $\mathbf{u} \in \text{Span}(S)$, then we have

$$\mathbf{u} = a_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} + a_2 \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} + a_3 \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ -(a_1 + a_2 + a_3) \end{bmatrix}$$

for some $a_1, a_2, a_3 \in \mathbb{R}$. Since $a_1 + a_2 + a_3 - (a_1 + a_2 + a_3) = 0, \mathbf{u} \in V$. Hence $\text{Span}(S) \subseteq V$.

Then let $\mathbf{w} \in V$,

$$\mathbf{w} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ -(x_1 + x_2 + x_3) \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \end{bmatrix}.$$

We have any $\mathbf{w} \in \text{Span}(S)$ and $V \subseteq \text{Span}(S)$. Thus $\text{Span}(S) = V$.

Consider the following augmented matrix for $S\mathbf{a} = \mathbf{0}$,

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & -1 & -1 & 0 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

The elements of S are linearly independent. Therefore S is a basis for V and the dimension of V is 3.

Question 7. In each of the following, find the dimension of the subspace of P_3 spanned by the given vectors:

(a) $x, x-1, x^2+1$

(b) $x, x-1, x^2+1, x^2-1$

(c) $x^2, x^2-x-1, x+1$

(d) $2x, x-2$

Solution

Take $x^2, x, 1$ as our standard ordered basis.

(a) Let's suppose that a_1, a_2, a_3 are constants such that:

$$\begin{aligned} a_1x + a_2(x-1) + a_3(x^2+1) &= 0 \\ \Rightarrow a_3x^2 + (a_1+a_2)x + (-a_2+a_3) &= 0 \end{aligned}$$

Put into matrix form:

$$\begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Thus, x, x_1, x^2+1 are linearly independent vectors, the dimension is **3**.

(b) Similarly, we can construct equation:

$$(a_3+a_4)x^2 + (a_1+a_2)x + (-a_2+a_3-a_4) = 0$$

Put into matrix and apply reduction:

$$\begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Thus, $x, x-1, x^2+1, x^2-1$ is linearly dependent vectors, the dimension is **3**.

(c) Similarly, we can construct equation:

$$(a_1 + a_2)x^2 + (-a_2 + a_3)x + (-a_2 + a_3) = 0$$

Put into matrix and apply reduction:

$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & -1 & 1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$
$$\Rightarrow \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Thus, $x^2, x^2 - x - 1, x + 1$ is linearly dependent vectors, the dimension is **2**.
(d) Similarly, we can construct equation:

$$(2a_1 + a_2)x + (-2a_2) = 0$$

Put into matrix and apply reduction:

$$\begin{bmatrix} 2 & 1 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$
$$\Rightarrow \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Thus, $2x, x - 2$ is linearly independent vectors, the dimension is **2**.

Question 8. Let $B = \left\{ \begin{bmatrix} 7 \\ 5 \end{bmatrix}, \begin{bmatrix} -3 \\ -1 \end{bmatrix} \right\}$, $C = \left\{ \begin{bmatrix} 1 \\ -5 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \end{bmatrix} \right\}$ be bases for $\mathbb{R}^2$. Find the transition matrix from B to C and the transition matrix from C to B.

Solution

Let P_{BC} and P_{CB} be transition matrix from B to C and C to B, then

$$P_{BC} = C^{-1}B = \begin{bmatrix} 1 & -2 \\ -5 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 7 & -3 \\ 5 & -1 \end{bmatrix} = \begin{bmatrix} -1/4 & -1/4 \\ -5/8 & -1/8 \end{bmatrix} \begin{bmatrix} 7 & -3 \\ 5 & -1 \end{bmatrix} = \begin{bmatrix} -3 & 1 \\ -5 & 2 \end{bmatrix}$$

and $P_{CB} = P_{BC}^{-1} = \begin{bmatrix} -3 & 1 \\ -5 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} -2 & 1 \\ -5 & 3 \end{bmatrix}$

Question 9. Let $b_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, b_2 = \begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix}, b_3 = \begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix}$.

(a) Show that the set $\mathbf{B} = \{b_1, b_2, b_3\}$ is a basis of $\mathbb{R}^3$.

(b) Given a vector $\mathbf{x} = \begin{bmatrix} -8 \\ 2 \\ 3 \end{bmatrix}$ in $\mathbb{R}^3$. Find the coordinate vector of $\mathbf{x}$ with respect to the basis $\mathbf{B}$.

(c) Given the coordinate of $\mathbf{y}$ with respect to the basis $\mathbf{B}$ is $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. What is the vector $\mathbf{y}$ in $\mathbb{R}^3$.

Solution

(a) We should prove the b_1, b_2, b_3 are linearly independent by using elementary row operations.

$$\begin{bmatrix} 1 & -3 & 3 \\ 0 & 4 & -6 \\ 0 & 0 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & -3 & 3 \\ 0 & 1 & -3/2 \\ 0 & 0 & 1 \end{bmatrix}$$

It is obvious that there are three pivot column. So b_1, b_2, b_3 are linearly independent and they can span the space $\mathbb{R}^3$. So $\mathbf{B} = b_1, b_2, b_3$ is a basis of $\mathbb{R}^3$.

(b) Let $c_1b_1 + c_2b_2 + c_3b_3 = x$, we should find $B[x]_B = x$ Apply elementary row operation on augmented matrix $[B|x]$.

$$\begin{bmatrix} 1 & -3 & 3 & -8 \\ 0 & 4 & -6 & 2 \\ 0 & 0 & 3 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & -3 & 3 & -8 \\ 0 & 1 & -3/2 & 1/2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 0 & -5 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Hence,

$$[x]_B = \begin{bmatrix} -5 \\ 2 \\ 1 \end{bmatrix}$$

(c) The vector y in $\mathbb{R}_3$, so $y = c_1b_1 + c_2b_2 + c_3b_3$. The scalars c_1, c_2 and c_3 are coordinates of y. $[y]_B = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$. So we can get: $y = 1\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + 2\begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix} + 3\begin{bmatrix} 3 \\ -6 \\ 3 \end{bmatrix} = \begin{bmatrix} 4 \\ -10 \\ 9 \end{bmatrix}$

Question 10. For each of the following choices of A and b, determine whether b is in the column space of A.

(a) $A = \begin{bmatrix} 3 & 6 \\ 1 & 2 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

(b) $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}, b = \begin{bmatrix} 2 \\ 5 \\ 2 \end{bmatrix}$

(c) $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \\ 1 & 2 \end{bmatrix}, b = \begin{bmatrix} 5 \\ 10 \\ 5 \end{bmatrix}$

Solution

(a) No.

$$\begin{bmatrix} 3 & 6 \\ 1 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

So $Col(A) = Span\left\{\begin{bmatrix} 3 \\ 1 \end{bmatrix}\right\}$

Hence $b = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \notin Col(A)$

(b) Yes. Because we can easily write $b = 5a_1 + 2 * a_2$

(c) Yes. We also can easily write that $b = a_1 + 2a_2$

Question 11. If $A = \begin{bmatrix} 1 & 2 & 3 & 5 & 0 & 2 & 4 \\ 2 & 1 & 3 & 4 & 7 & 8 & 9 \\ 1 & 1 & 2 & 3 & 2 & 2 & 5 \end{bmatrix}$

(a) Find a basis for $Col(A)$

(b) Find a basis for $Null(A)$ and find the dimension of $Null(A)$.

Solution

(a) We do RREF can get the matrix:

$$\begin{bmatrix} 1 & 2 & 3 & 5 & 0 & 2 & 4 \\ 2 & 1 & 3 & 4 & 7 & 8 & 9 \\ 1 & 1 & 2 & 3 & 2 & 2 & 5 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & -14 & 14 \\ 0 & 1 & 1 & 2 & 0 & 8 & -5 \\ 0 & 0 & 0 & 0 & 1 & 4 & -2 \end{bmatrix}$$

So the pivot columns of B are $\{1, 2, 5\}$. Hence the basis for $Col(A)$ is $\left\{ \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 7 \\ 2 \end{bmatrix} \right\}$

(b) From the RREF matrix of (a), we can write:

$$\begin{cases} x_1 = -x_3 - x_4 + 14x_6 - 14x_7 \\ x_2 = -x_3 - 2x_4 - 8x_6 + 5x_7 \\ x_5 = -4x_6 + 2x_7 \end{cases}$$

So the basis of $Null(A)$ is

$$\left\{ \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 14 \\ -8 \\ 0 \\ 0 \\ -4 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -14 \\ 5 \\ 0 \\ 0 \\ 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Hence the dimension of $Null(A)$ is 4.

Question 12. Let $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ be linearly independent vectors in $\mathbb{R}^n$ and let

$$\mathbf{y}_1 = \mathbf{x}_2 - \mathbf{x}_1, \mathbf{y}_2 = \mathbf{x}_3 - \mathbf{x}_2, \mathbf{y}_3 = \mathbf{x}_3 - \mathbf{x}_1$$

Are $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3$ linearly independent? Prove your answer.

Solution

If $c_1\mathbf{y}_1 + c_2\mathbf{y}_2 + c_3\mathbf{y}_3 = 0$, then

$$c_1(\mathbf{x}_2 - \mathbf{x}_1) + c_2(\mathbf{x}_3 - \mathbf{x}_2) + c_3(\mathbf{x}_3 - \mathbf{x}_1) = \mathbf{0}$$

Rearranging terms

$$(-c_1 - c_3)\mathbf{x}_1 + (c_1 - c_2)\mathbf{x}_2 + (c_2 + c_3)\mathbf{x}_3 = \mathbf{0}$$

Since $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are linearly independent, the coefficients in equation above must all be 0. Thus we have the linear system

$$\begin{aligned} -c_1 - c_3 &= 0 \\ c_1 - c_2 &= 0 \\ c_2 + c_3 &= 0 \end{aligned}$$

To solve, we reduce to row echelon form

$$\left[\begin{array}{ccc|c} -1 & 0 & -1 & 0 \\ 1 & -1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Since the row echelon form involves free variables, there will be nontrivial solutions. Since there are nontrivial linear combinations of $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3$ that equals to 0, the vectors must be linearly dependent.

Question 13. Let

$$A = \begin{bmatrix} 1 & 2 & 2 & 3 & 1 & 4 \\ 2 & 4 & 5 & 5 & 4 & 9 \\ 3 & 6 & 7 & 8 & 5 & 9 \end{bmatrix}$$

Compute the reduced row echelon form U of A. Which column vectors of A correspond to the dependent variables of U? These column vectors form a basis for the column space of A. Write each of the remaining column vectors of A as a linear combination of these basis vectors.

Solution

The reduced row echelon form of A is given by

$$U = \begin{bmatrix} 1 & 2 & 0 & 5 & -3 & 0 \\ 0 & 0 & 1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

The dependent variables correspond to columns 1, 3, and 6. Thus $\mathbf{a}_1$, $\mathbf{a}_3$, $\mathbf{a}_6$ form a basis for the column space of A. The remaining column vectors satisfy the following dependency relationships.

$$\mathbf{a}_2 = 2\mathbf{a}_1$$

$$\mathbf{a}_4 = 5\mathbf{a}_1 - \mathbf{a}_3$$

$$\mathbf{a}_5 = -3\mathbf{a}_1 + 2\mathbf{a}_3$$

Question 14. Let A be a 4×4 matrix with reduced row echelon form given by

$$U = \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 1 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

If

$$\mathbf{a}_1 = \begin{bmatrix} -3 \\ 5 \\ 2 \\ 1 \end{bmatrix} \quad \text{and} \quad \mathbf{a}_1 = \begin{bmatrix} 4 \\ -3 \\ 7 \\ -1 \end{bmatrix}$$

find $\mathbf{a}_3$ and $\mathbf{a}_4$.

Solution

The column vectors of A and U satisfy the same dependency relations. By inspection one can see that

$$\mathbf{u}_3 = 2\mathbf{u}_1 + \mathbf{u}_2 \quad \text{and} \quad \mathbf{u}_4 = \mathbf{u}_1 + 4\mathbf{u}_2$$

Therefore

$$\mathbf{a}_3 = 2\mathbf{a}_1 + \mathbf{a}_2 = \begin{bmatrix} -6 \\ 10 \\ 4 \\ 2 \end{bmatrix} + \begin{bmatrix} 4 \\ -3 \\ 7 \\ -1 \end{bmatrix} = \begin{bmatrix} -2 \\ 7 \\ 11 \\ 1 \end{bmatrix}$$

and

$$\mathbf{a}_4 = \mathbf{a}_1 + 4\mathbf{a}_2 = \begin{bmatrix} -3 \\ 5 \\ 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 16 \\ -12 \\ 28 \\ -4 \end{bmatrix} = \begin{bmatrix} 13 \\ -7 \\ 30 \\ -3 \end{bmatrix}$$

Question 15. Let A be a 4×3 matrix and suppose that the vectors

$$\mathbf{z}_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad \mathbf{z}_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

form a basis for $N(A)$, If $\mathbf{b} = \mathbf{a}_1 + 2\mathbf{a}_2 + \mathbf{a}_3$, find all solutions of the system $A\mathbf{x} = \mathbf{b}$.

Solution

Suppose $A = [\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3]$, where $\mathbf{a}_i (i = 1, 2, 3)$ are column vectors of A.

Since $\mathbf{z}_1, \mathbf{z}_2$ are a set of basis that forms $N(A)$, its means that $\mathbf{z} = c_1\mathbf{z}_1 + c_2\mathbf{z}_2$ $(\forall c_1, c_2\ \in \mathbb{R})$ solves the system of $A\mathbf{z} = \mathbf{0}$, i.e.,

$$A\mathbf{z} = \begin{bmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 \end{bmatrix} \begin{bmatrix} c_1 + c_2 \\ c_1 \\ 2c_1 - c_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

which gives $(c_1 + c_2)\,\mathbf{a}_1 + c_1\mathbf{a}_2 + (2c_1 - c_2)\,\mathbf{a}_3 = \mathbf{0}$, denoted as Eq.1.

Denote $\mathbf{x} = [x_1\ x_2\ x_3]^\top$,

$$A\mathbf{x} = [\mathbf{a}_1\ \mathbf{a}_2\ \mathbf{a}_3] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + x_3\mathbf{a}_3 = \mathbf{a}_1 + 2\mathbf{a}_2 + \mathbf{a}_3 = \mathbf{b},$$

which gives $(x_1 - 1)\,\mathbf{a}_1 + (x_2 - 2)\,\mathbf{a}_2 + (x_3 - 1)\,\mathbf{a}_3 = \mathbf{0}$, denoted as Eq.2.

Comparing coefficients of Eq.1 and Eq.2, we have

$$\begin{cases} c_1 + c_2 = x_1 - 1 \\ c_1 = x_2 - 2 \\ 2c_1 - c_2 = x_3 - 1 \end{cases}, \text{ whose solution is: } \begin{cases} x_1 = c_1 + c_2 + 1 \\ x_2 = c_1 + 2 \\ x_3 = 2c_1 - c_2 + 1 \end{cases}$$

Therefore,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = c_1\mathbf{z}_1 + c_2\mathbf{z}_2 + \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}, \forall c_1, c_2\ \in \mathbb{R}$$

Question 16. Suppose $(x_1, x_2) + (y_1, y_2)$ is defined to be $(x_1 + y_2, x_2 + y_1)$. With the usual multiplication $c\boldsymbol{x} = (cx_1, cx_2)$, which of the eight axioms of a vector space (see Lecture 11) are **not** satisfied?

Solution

$(x_1, x_2) + (y_1, y_2) \neq (y_1, y_2) + (x_1, x_2) \Rightarrow$ A1 is not satisfied.

$[(x_1, x_2) + (y_1, y_2)] + (z_1, z_2) + \neq (x_1, x_2) + [(y_1, y_2) + (z_1, z_2)] \Rightarrow$ A2 is not satisfied.

(x1, x2) + (-x1, -x2) is not 0, A4 is not satisfied.

$(c_1 + c_2)(x_1, x_2) \neq c_1(x_1, x_2) + c_2(x_1, x_2) \Rightarrow$ A7 is not satisfied.

Question 17. Find a basis for the following solution set:

$$\left\{ \mathbf{x} \in \mathbb{R}^4 \middle| \begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix} \mathbf{x} = 0 \right\}$$

Solution

Let $A = \begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix}$, we can do the elementary row operations to get the echolon form.

$$\begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix} \to \begin{bmatrix} 1 & -2 & 1 & 2 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix} \to$$

$$\begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & -2 & -5 & -3 \\ 0 & 2 & 5 & 3 \end{bmatrix} \to \begin{bmatrix} 1 & -2 & 1 & 2 \\ 0 & 2 & 5 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \to \begin{bmatrix} 1 & 0 & 6 & 5 \\ 0 & 1 & 5/2 & 3/2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

So from the last matrix, we know the x_3 and x_4 are free variables, and $x_1 = -6x_3 - 5x_4$, $x_2 = -5/2x_3 - 3/2x_4$.

So we can write the result:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -6 \\ -5/2 \\ 1 \\ 0 \end{bmatrix} x_3 + \begin{bmatrix} -5 \\ -3/2 \\ 0 \\ 1 \end{bmatrix} x_4$$

So one of the basis is: $\begin{bmatrix} -6 \\ -5/2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ -3/2 \\ 0 \\ 1 \end{bmatrix}$

Question 18. Let $\{x, 1\}$ and $\{2x-1, 2x+1\}$ be ordered bases for P_2, find the transition matrix representing the change in coordinates from $\{2x-1, 2x+1\}$ to $\{x, 1\}$

Solution

Since

$$2x - 1 = 2 \cdot x + (-1) \cdot 1$$
$$2x + 1 = 2 \cdot x + 1 \cdot 1$$

Let transition matrix $A = [a_1, a_2]$, then $a_1 = [2, -1]^T$, $a_2 = [2, 1]^T$

$$A = \begin{bmatrix} 2 & 2 \\ -1 & 1 \end{bmatrix}$$

Question 19. Find a basis for following matrix:

$$\text{Null}\left(\begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix}\right)$$

Solution

Let

$$A = \begin{bmatrix} -2 & 4 & -2 & -4 \\ 2 & -6 & -3 & 1 \\ -3 & 8 & 2 & -3 \end{bmatrix}$$

We can get the reduced echelon form as:

$$B = \begin{bmatrix} 1 & 0 & 6 & 5 \\ 0 & 1 & 2.5 & 1.5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

so $\text{Null}(A) = \{x \in \mathbb{R}^n : Bx = 0\} = Span\left\{\begin{bmatrix} -6 \\ -2.5 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -5 \\ -1.5 \\ 0 \\ 1 \end{bmatrix}\right\}$

Question 20. For each of the following matrices, find a basis for the row space, a basis for the column space, and a basis for the null space.

(a)

$$\begin{bmatrix} 1 & 3 & 2 \\ 2 & 1 & 4 \\ 4 & 7 & 8 \end{bmatrix}$$

(b)

$$\begin{bmatrix} -3 & 1 & 3 & 4 \\ 1 & 2 & -1 & -2 \\ -3 & 8 & 4 & 2 \end{bmatrix}$$

(c)

$$\begin{bmatrix} 1 & 3 & -2 & 1 \\ 2 & 1 & 3 & 2 \\ 3 & 4 & 5 & 6 \end{bmatrix}$$

Solution

(a) The reduced row echelon form of the matrix is

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Thus (1, 0, 2) and (0, 1, 0) form a basis for the row space. The first and second columns of the original matrix form a basis for the column space:

$$\mathbf{a}_1 = (1,\, 2,\, 3)^T,\ \mathbf{a}_2 = (3,\, 1,\, 7)^T$$

The reduced row echelon form involves one free variable and hence the nullspace will have dimension 1. Setting $x_3 = 1$, we get $x_1 = -2$ and $x_2 = 0$. Thus $(-2,\, 0,\, 1)^T$ is a basis for the nullspace.

(b) The reduced row echelon form of the matrix is

$$\begin{bmatrix} 1 & 0 & 0 & -10/7 \\ 0 & 1 & 0 & -2/7 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

Clearly then, the set

$$(1,\, 0,\, 0,\, -10/7),\ (0,\, 1,\, 0,\, -2/7),\ (0,\, 0,\, 1,\, 0)$$

is a basis for the row space. Since the reduced row echelon form of the matrix involves one free variable the nullspace will have dimension 1. Setting the free variable $x_4 = 1$ we get

$$x_1 = 10/7,\ x_2 = 2/7,\ x_3 = 0$$

Thus $\{(10/7,\, 2/7,\, 0,\, 1)^T\}$ is a basis for the nullspace. The dimension of the column space equals the rank of the matrix which is 3. Thus the column space must be R^3 and we can take as our basis the standard basis $\{\mathbf{e}_1,\, \mathbf{e}_2,\, \mathbf{e}_3\}$.

(c) The reduced row echelon form of the matrix is

$$\begin{bmatrix} 1 & 0 & 0 & -0.65 \\ 0 & 1 & 0 & 1.05 \\ 0 & 0 & 1 & 0.75 \end{bmatrix}$$

The set $\{(1, 0, 0, -0.65), (0, 1, 0, 1.05), (0, 0, 1, 0, 0.75)\}$ is a basis for the row space. The set $\{(0.65, -1.05, -0.75, 1)^T\}$ is a basis for the nullspace. As in part (b) the column space is R3 and we can take $\{\mathbf{e}_1,\, \mathbf{e}_2,\, \mathbf{e}_3\}$ as our basis.